



Mill Test Certificate

Inspection certificate, according to EN 10204, type 3.1

Trade name: EOS NickelAlloy IN625 (EOS art.-no.: 9011-0022)

Lot number: M421601

Manufacturer: Electro Optical Systems Finland Oy
Lemminkäisenkatu 36
FI - 20520 Turku
FINLAND

Supplier: EOS GmbH
Robert-Stirling-Ring 1
D-82152 Krailling Germany
Telephone: +49 (0)89 / 893 36-0
Telefax: +49 (0)89 / 893 36-285

Customs tariff code: 7504 0000 00

Analyses of powder

Property	Test method	Units	Specification	Result
Particle size analysis		Upper Limit		
	Sieve analysis > 53µm	%	3.0	1.0
	Sieve analysis > 63µm	%	0.5	0.1



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Analyses of powder

Property	Analysis method	Analysis Standard	Specifica- tion	Specifica- tion	Result
Chemical composition			Lower Limit	Upper Limit	
Ni	ICP	ASTM E2823/E1479	58.00	Balance	64.66
Cr	ICP	ASTM E2823/E1479	20.00	23.00	20.61
Fe	ICP	ASTM E2823/E1479	-	5.00	0.81
Mo	ICP	ASTM E2823/E1479	8.00	10.00	8.82
Nb	ICP	ASTM E2823/E1479	3.15	4.15	3.97
Co	ICP	ASTM E2823/E1479	-	1.00	0.17
Ti	ICP	ASTM E2823/E1479	-	0.40	0.39
Al	ICP	ASTM E2823/E1479	-	0.40	0.30
Si	ICP	ASTM E2823/E1479	-	0.50	0.18
Mn	ICP	ASTM E2823/E1479	-	0.50	0.04
P	ICP	ASTM E2823/E1479	-	0.015	<0.010
Ta	ICP	ASTM E2823/E1479	-	0.05	0.01
C	Combustion	ASTM E1019	-	0.10	0.02
S	Combustion	ASTM E1019	-	0.015	<0.005
O	Fusion	ASTM E1019	-	-	0.013
N	Fusion	ASTM E1019	-	-	0.012



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Properties of laser processed EOS NickelAlloy IN625 material

Following properties have been determined on laser processed EOS NickelAlloy IN625 lot M421601 when processed in following setup:

- EOSINT M290 machine
- IN625_040_PerformanceM291_110 default parameters

Property	Test method	Units	Specification	Result
Density	Weighing in air and water acc. to ISO 3369	g/cm ³	≥ 8.4	8.5
Tensile properties in heat treated condition* <i>Vertical orientation</i>	Tensile testing acc. to DIN EN ISO 6892-1:2009 B10			
Yield strength R _{p0.2%}		MPa	≥ 414	630
Tensile strength R _m		MPa	≥ 827	915
Percentage elongation after fracture A ₂₅		%	≥ 30	42
Hardness in heat treated condition*	Hardness testing acc. to EN ISO 6508-1	HRC	-	29

* Heat treatment: Stress relieve anneal at 870°C for 1 hour, rapid cooling.

Specimen for tensile testing: cylindric, d=5 mm



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This document confirms that the material defined above has been tested and is in compliance with the product specification for use in EOSINT M Systems for direct metal laser sintering following the operation instructions. The data correspond to our knowledge and experience at the time of publication. They do not on their own represent a sufficient basis for any part design, neither do they provide any agreement about or guarantee the specific properties of a product or part or the suitability of a product or part for a specific application. It is the responsibility of the producer or customer of a part to check its properties as well as its suitability for a particular purpose. This also applies regarding the consideration of possible intellectual property rights as well as laws and regulations. The data are subject to change without notice as part of EOS' continuous development and improvement processes. EOS has not FDA cleared this product for medical device manufacturers to use this material in FDA sensitive applications.

Date:

30.11.2016

Approved by:

Toni Kosunen

Toni Kosunen

Quality Engineer

Electro Optical Systems Finland Oy